EMPLOYEE SALARY PREDICTION

DATA SCIENCE PROJECT REPORT

Random Forest Regression • Exploratory Data Analysis • Model Evaluation

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1. Project Overview

Goal: estimate annual salary in USD using job and employment details.

Dataset summary	Value
File	employee_salary(1).csv
Original records	11,087
Columns	11
Exact duplicate rows removed	4,488
Prediction target	salary_in_usd

2. Exploratory Data Analysis (EDA)

Graphs highlight salary spread, experience-level differences, common job titles, and company-size patterns.

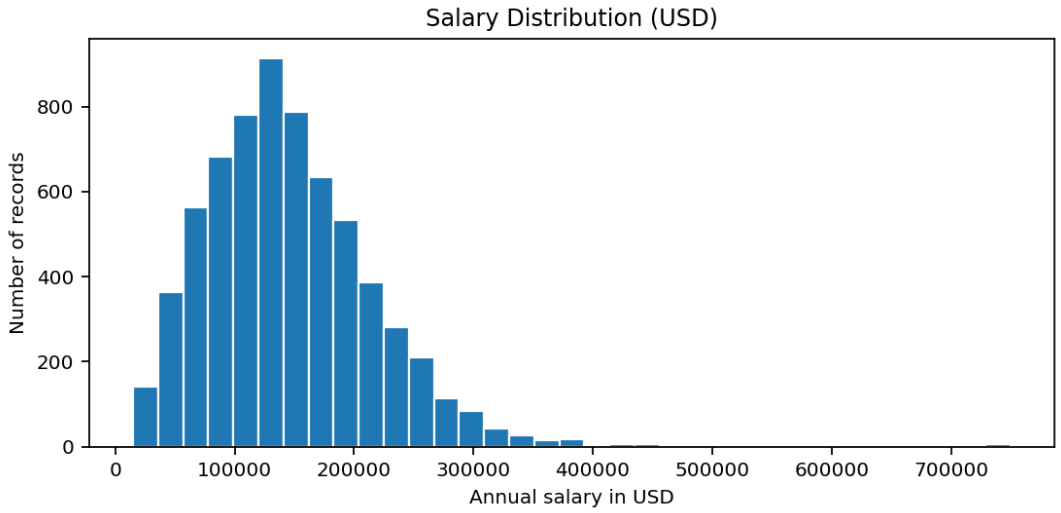


Figure 1. Salary distribution is right-skewed; a small number of high salaries extend the range.

EDA: Key Patterns

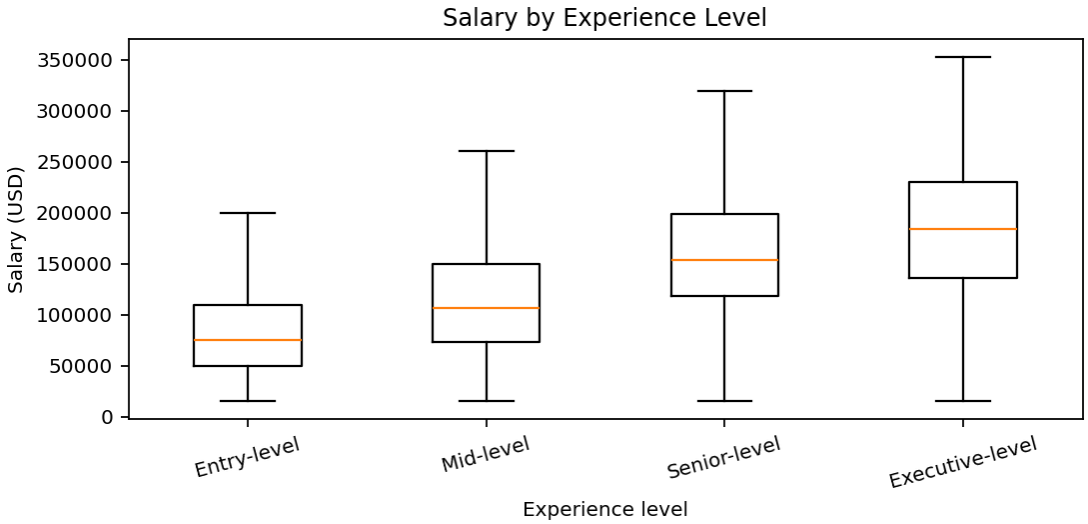


Figure 2. Salary patterns across experience levels.

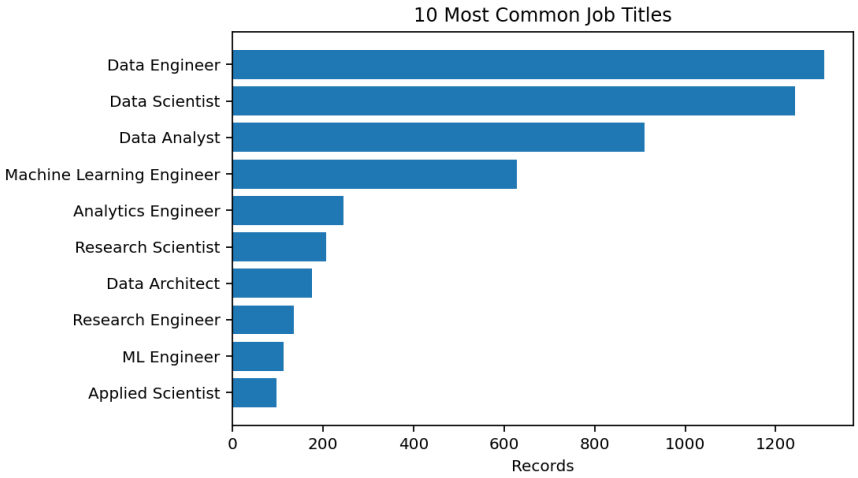


Figure 3. Most frequent job titles in the dataset.

3. Data Preparation

Step	What was done
Cleaning	Removed exact duplicate rows and records with missing target salary.
Outlier detection	Used the IQR rule on salary_in_usd. Values outside Q1 – 1.5×IQR and Q3 + 1.5×IQR were flagged.
Outlier removal	107 records (1.62%) were removed; 6,492 rows remained for the reported modelling workflow.
Encoding	OneHotEncoder converts categorical features into numeric indicator columns. Unknown categories are ignored.
Scaling / standardization	StandardScaler transforms work_year to mean 0 and standard deviation 1. It is fitted within the training pipeline.

IQR outlier rule

Q1 = $95,000 • Q3 = $185,000 • IQR = $90,000

Lower fence = –$40,000 • Upper fence = $320,000

Note: high salaries may be valid observations. Outlier removal changes the dataset and should be compared with a model trained without filtering.

4. Model Training & Evaluation

Model: Random Forest Regressor

Setting	Value
Train/test split	80% / 20%
Random state	42
Trees	100
Test examples	1,299
R^2 score	0.391

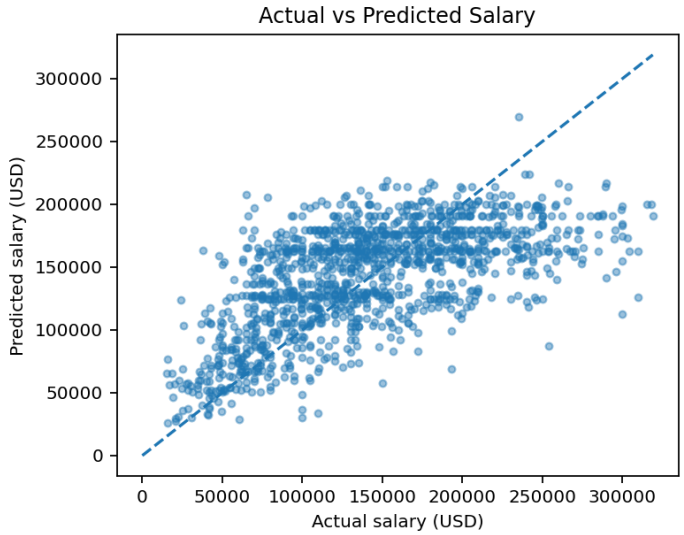
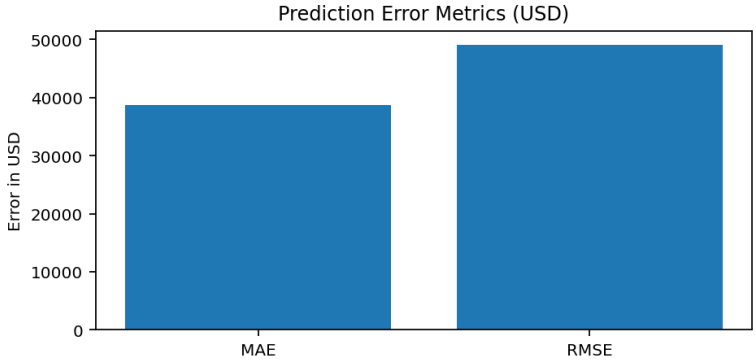


Figure 4. Actual vs. predicted salaries on the test set. Points closer to the diagonal indicate better predictions.



MAE, RMSE and MSE summarize prediction error; lower values are better.

5. Findings & Conclusion

Key findings

• Salary varies by experience, job title, and company context.
• Test $R^2 = 0.391$: the model explains a limited share of salary variation.
• Salary also depends on skills, responsibilities, industry, and local labour markets.

Limitations

• Dataset categories may be imbalanced.
• IQR filtering can remove genuine high salaries.
• Job titles and locations create many encoded features.
• Historical patterns may not generalize to future salaries.

Future improvements

• Compare with a median-salary baseline and other regressors.
• Use cross-validation and hyperparameter tuning.
• Compare results with and without outlier removal.
• Validate prediction inputs against dataset labels (e.g., “Senior-level”, “Full-time”).
• Add feature importance and error analysis by experience and location.

Conclusion

This project demonstrates a complete salary-prediction workflow: cleaning, EDA, outlier handling, encoding, standardization, model training, and evaluation. The model is a learning baseline, not a guaranteed salary estimate.

Dataset source URL was not verified from the supplied notebook/CSV.